

Control work 1, and work on the mistakes

One hour of written assessment on the revision block and the definition of a function; one hour putting right what it exposes.

LESSON 15–16

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 1

P1 · CHAPTER 1–2 REVIEW

LESSON CLOCK

3

37'

10'

25'

5'

Instructions	3 min
The paper	37 min
Self-mark and classify	10 min
Rewrite the wrong ones	25 min
Set the resit list	5 min

LEARNING OBJECTIVES

- Work accurately and independently under time.
- Show the reasoning, not only the answer.
- Classify each mistake as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 48–49
Cambridge	Pure Mathematics 1, pp. 24, 42
Workbook	Control paper A

01

Explanation

The paper — 20 marks, 37 minutes

Q	TASK	MARKS	FROM
1	Solve $x^2 - 7x + 10 = 0$ by factorising	2	L2–3
2	Solve $3x^2 - 5x - 2 = 0$ by formula, exact answers	3	L2–3
3	Write $x^2 - 8x + 21$ in completed-square form and give the vertex	3	L4–5
4	Classify three mappings and say which are functions	3	L8
5	$f(x) = x^2 - 3x$: find $f(-2)$ and $f(a + 1)$	3	L9–10
6	Domain and range of $f(x) = \sqrt{x - 5}$	3	L11–12
7	$f = 2x, g = x - 4$: find $(f/g)(x)$ and its domain	3	L13–14

MARKING

A correct answer with no working scores half. A wrong answer reached by correct method scores most of the marks. Write the method.

Classifying your own mistakes

In the second hour, every learner rules three columns and puts each lost mark in one:

KIND	LOOKS LIKE	WHAT FIXES IT
careless	a sign, a copied digit, an unfinished last line	checking, not revision
method	the right idea applied in the wrong order	one more worked example
knowledge	you did not know the fact or the rule	re-read the section, then five problems

THE POINT OF THE EXERCISE

A learner with six careless marks and a learner with six knowledge marks scored the same and need completely different homework. That is why the column matters more than the total.

The rewrite

Every question that lost a mark is rewritten in full — not corrected in the margin.

Under each rewrite, one sentence: *"I lost this because ..."*.

$$x^2 + 6x + 9 = (x + 3)^2$$

$$\text{so } x^2 + 6x = (x + 3)^2 - 9$$

Question 3 in one picture — the square, the correction, the vertex.

Learners with more than four knowledge marks in one section get that section's problem set as their homework instead of the standard list.

02 Worked examples

EXAMPLE 1 Model answer, Q3: write $x^2 - 8x + 21$ in completed-square form.

1 Half of -8 is -4 .
 $(x - 4)^2 = x^2 - 8x + 16$

2 We need $+21$, we have $+16$.
 Add 5.

3 $x^2 - 8x + 21 = (x - 4)^2 + 5$
 Vertex $(4, 5)$.

ANSWER

$(x - 4)^2 + 5$, vertex $(4, 5)$

EXAMPLE 2 Model answer, Q7: $f = 2x$, $g = x - 4$. Find $(f/g)(x)$ and its domain.

1 $(f/g)(x) = \frac{2x}{x - 4}$

2 Both domains are $\mathbb{R}$.

3 Remove the zero of g : $x = 4$.

ANSWER

$\frac{2x}{x - 4}$, domain $x \neq 4$

EXAMPLE 3 A learner wrote $f(a + 1) = f(a) + 1 = a^2 - 3a + 1$. Name the mistake.

- ① They added 1 to the output instead of the input.
A method error, not carelessness.

- ② Correct: $(a + 1)^2 - 3(a + 1)$

- ③ $= a^2 + 2a + 1 - 3a - 3 = a^2 - a - 2$

ANSWER

Method error; the answer is $a^2 - a - 2$

03 Interactive model

Re-do the two questions the class lost most marks on before the rewrite starts.

Mark these six answers

$x^2 - 7x + 10 = (x - 2)(x - 5)$ — is it right?

yes

no, sign error

no, wrong numbers

no, it does not factorise

For $3x^2 - 5x - 2$, $D = 25 - 24 = 1$. Right?

yes

no — $D = 49$

no — $D = 1$ but sign wrong

no — $D = -49$

$x^2 - 8x + 21 = (x - 4)^2 + 21$

right

careless

method error — forgot to subtract 16

knowledge gap

Domain of $\sqrt{x-5}$ given as $x > 5$

right

careless — should be $x \geq 5$

method error

right, both work

$(f/g)(x) = \frac{2x}{x-4}$ with domain $\mathbb{R}$

right

careless

method — forgot $x \neq 4$

knowledge gap

$x \mapsto \pm\sqrt{x}$ called a function

right

knowledge gap

careless

method error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Assessment criteria	Baholash mezonlari	Критерии оценивания
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Working (shown)	Yechim yozuvi	Ход решения
Mark scheme	Baholash sxemasi	Схема оценивания
Correction	Tuzatish	Исправление
Resit task	Qayta topshiriq	Повторное задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A correct answer with no working scores:

full marks

half

nothing

one mark

A dropped minus sign is best classed as:

careless

method

knowledge

unavoidable

Not knowing what a discriminant is, is:

careless

method

knowledge

careless

The most useful number from a control work is:

- the total
- the class average
- the column your losses fell in
- the rank

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x^2 - 7x + 10 = 0$

ANSWER $x = 2, 5$

2. Solve $x^2 - 9 = 0$

ANSWER $x = \pm 3$

3. Complete the square: $x^2 + 2x$

ANSWER $(x + 1)^2 - 1$

4. $f(x) = x^2 - 3x$; find $f(-2)$

ANSWER 10

5. Domain of $\sqrt{x - 5}$

ANSWER $x \geq 5$

6. Is $x \mapsto x^2$ a function?

ANSWER yes

7. $f = 2x$, $g = x - 4$; find $(f + g)(5)$

ANSWER 11

Medium · the standard question

7 PROBLEMS

1. Solve $3x^2 - 5x - 2 = 0$

ANSWER $x = 2, -\frac{1}{3}$

2. Complete the square: $x^2 - 8x + 21$

ANSWER $(x - 4)^2 + 5$

3. $f(x) = x^2 - 3x$; find $f(a + 1)$

ANSWER $a^2 - a - 2$

4. Range of $y = (x - 4)^2 + 5$

ANSWER $y \geq 5$

5. Domain of $\frac{2x}{x - 4}$

ANSWER $x \neq 4$

6. Classify $x \mapsto x^2 + 1$ on $\mathbb{R}$

ANSWER many-to-one

7. Solve $2x^2 + 7x + 3 = 0$

ANSWER $x = -3, -0.5$

Hard · stretch and reason

7 PROBLEMS

1. For what k does $x^2 + kx + 16 = 0$ have one root?

ANSWER $k = \pm 8$

2. Range of $y = x^2 - 8x + 21$ on $2 \leq x \leq 5$

ANSWER $5 \leq y \leq 9$

3. $f(x) = x^2$; simplify $\frac{f(x+h) - f(x)}{h}$

ANSWER $2x + h$

4. Domain of $\frac{\sqrt{x-1}}{x-3}$

ANSWER $x \geq 1, x \neq 3$

5. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

6. Find a, b with $x^2 + ax + b = (x-3)^2 - 4$

ANSWER $a = -6, b = 5$

7. Explain why $D < 0$ means no x -intercepts

ANSWER The parabola never reaches the axis

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every question that lost a mark, with the “I lost this because ...” sentence.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $2x^2 - 9x + 4 = 0$ twice — factorising and by formula — and check they agree.

4. Find the domain and range of $f(x) = \frac{3}{x+1}$.